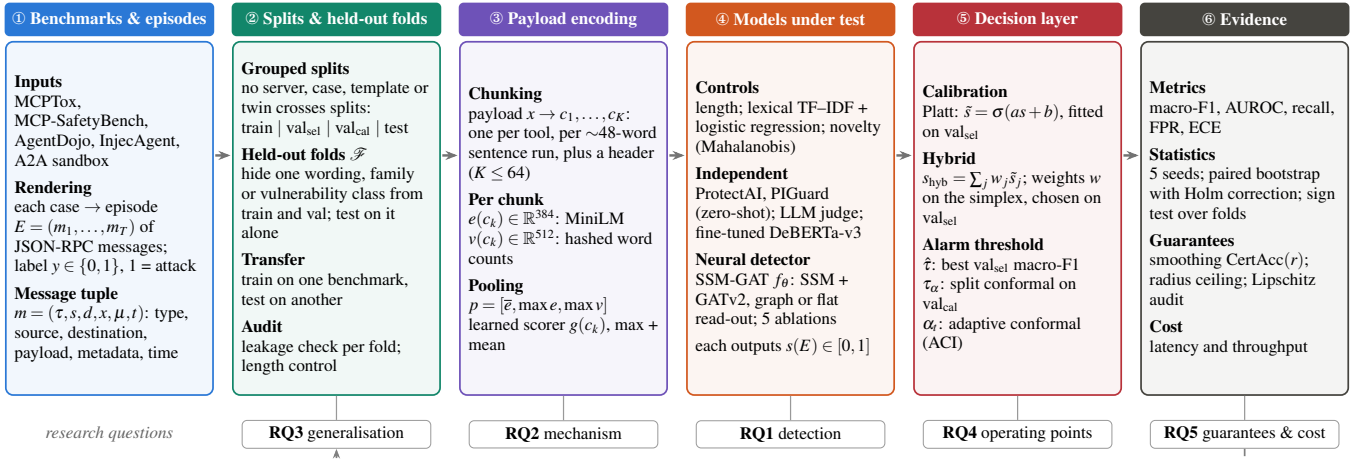

① Benchmarks & episodes
Inputs
MCPTox, MCP-SafetyBench, AgentDojo, InjecAgent, A2A sandbox
Rendering
each case → episode $E = (m_1, \ldots, m_T)$ of JSON-RPC messages; label $y \in \{0, 1\}$, 1 = attack
Message tuple
$m = (\tau, s, d, x, \mu, t)$: type, source, destination, payload, metadata, time
② Splits & held-out folds
Grouped splits
no server, case, template or twin crosses splits: train | val_{sel} | val_{cal} | test
Held-out folds $\mathscr{F}$
hide one wording, family or vulnerability class from train and val; test on it alone
Transfer
train on one benchmark, test on another
Audit
leakage check per fold; length control
③ Payload encoding
Chunking
payload $x \to c_1, \ldots, c_K$: one per tool, per ∼48-word sentence run, plus a header ($K \leq 64$)
Per chunk
$e(c_k) \in \mathbb{R}^{384}$: MiniLM
$v(c_k) \in \mathbb{R}^{512}$: hashed word counts
Pooling
$p = [\bar{e}, \max e, \max v]$ learned scorer $g(c_k)$, max + mean
④ Models under test
Controls
length; lexical TF–IDF + logistic regression; novelty (Mahalanobis)
Independent
ProtectAI, PIGuard (zero-shot); LLM judge; fine-tuned DeBERTa-v3
Neural detector
SSM-GAT f_θ: SSM + GATv2, graph or flat read-out; 5 ablations
each outputs $s(E) \in [0, 1]$
⑤ Decision layer
Calibration
Platt: $\tilde{s} = \sigma(as + b)$, fitted on val_{sel}
Hybrid
$s_{\text{hyb}} = \sum_j w_j \tilde{s}_j$; weights w on the simplex, chosen on val_{sel}
Alarm threshold
$\hat{\tau}$: best val_{sel} macro-F1
τ_α: split conformal on val_{cal}
α_t: adaptive conformal (ACI)
⑥ Evidence
Metrics
macro-F1, AUROC, recall, FPR, ECE
Statistics
5 seeds; paired bootstrap with Holm correction; sign test over folds
Guarantees
smoothing CertAcc(r); radius ceiling; Lipschitz audit
Cost
latency and throughput
research questions
RQ3 generalisation
RQ2 mechanism
RQ1 detection
RQ4 operating points
RQ5 guarantees & cost
pre-registered: every choice uses val_{sel} only; val_{cal} only sets conformal thresholds; the test split is read once